



Genomic characterisation of *Pseudomonas asiatica* as an emerging mastitis pathogen in dairy cows with resistance and virulence implications

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ABSTRACT

Objective: Mastitis, a major concern in the dairy industry, is often caused by multidrug-resistant (MDR) pathogens, making treatment more challenging. This study aimed to elucidate the prevalence, antimicrobial resistance (AMR), antibiotic resistance genes (ARGs) and virulence genes (VGs) profiles of *Pseudomonas asiatica* strains isolated from milk, faeces, and farm soils of dairy cows diagnosed with clinical mastitis (CM).

Methods: Forty-five milk and 35 faeces samples from cows with CM, as well as 30 soil samples were collected from 30 small-holding dairy farms in Gazipur district, Bangladesh. *P. asiatica* was identified using culture-based methods, VITEK-2, and 16S rRNA sequencing. AMR profile was determined using disc diffusion method. Whole-genome sequencing (WGS) of four MDR *P. asiatica* isolates was performed for phylogenetic analysis, ARGs and VGs prediction.

Results: The overall prevalence of *P. asiatica* in the investigated dairy farms in this study was 37.27%, with rates of 29.27% in milk, 34.15% in faeces, and 36.59% in soil. All *P. asiatica* isolates were MDR, exhibiting highest resistance to imipenem, ampicillin, oxacillin, sulfonamides, nitrofurantoin, azithromycin, tetracycline, and cefoxitin. WGS revealed close phylogenetic ties between *P. asiatica* isolates (2M1, 2F1, 2F2, 2S1) and global *Pseudomonas* strains. Each of the four *P. asiatica* strains harbored 11 ARGs imparting resistance to multiple antibiotics, and 30 VGs linked to motility, alginate/pyoverdine synthesis, and regulatory systems.

Conclusions: This study identifies *P. asiatica* as an emerging MDR mastitis pathogen with significant environmental persistence. These findings highlight for species-specific diagnostics, and tailored antimicrobial strategies to improve mastitis control in dairy farming systems.

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1. Introduction

Mastitis is one of the most common and devastating disease encountered in dairy farming and annually incurs substantial economic losses in the dairy industry, primarily caused by a broad

range of bacterial pathogens. Mastitis is a multifactorial disease caused by a diverse array of etiological agents [1–3]. *Pseudomonas* spp. are major contributors to bovine mastitis [4], where they employ diverse pathogenic mechanisms involving multiple virulence genes [5]. In addition, *Pseudomonas* spp. possess zoonotic potential, with the ability to be transmitted from animals to humans, causing respiratory infections, urinary tract infections, bloodstream infections, and ear infections [6,7]. While *Pseudomonas aeruginosa*, *P. fluorescens*, and *P. putida* are widely considered as predominantly opportunistic environmental pathogens associated with bovine mastitis [8,9], we previously identified 16 distinct strains of *Pseudomonas* spp. linked to CM in dairy cows [10]. This finding un-

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underscores the genetic and phenotypic diversity within the *Pseudomonas* genus and the varying pathogenic potential among different strains [5,10]. Importantly, the treatment of *Pseudomonas*-associated mastitis is difficult because the bacteria can acquire ARGs, making common antibiotics less effective [4,5]. Furthermore, antimicrobial resistance (AMR) in mastitis pathogens represents a significant public health risk, as resistant bacteria and antibiotic residues can be transmitted to humans, particularly through the consumption of raw milk from treated cows [11]. The global burden of AMR is growing, with more than 1.2 million direct deaths linked to AMR in 2019. Projections indicate that by 2050, AMR could cause approximately 10 million deaths annually if effective interventions are not implemented [12]. *P. asiatica*, a recently identified species within the *P. putida* group, is phylogenetically closely related to *P. putida* and *P. monteilii*. It has been isolated from hospitalized patients in Japan and Myanmar, indicating its potential clinical relevance [6,7]. As an emerging opportunistic pathogen, *P. asiatica* holds significance in both human and animal health. In humans, it has been recovered from various clinical samples, including stool from patients with diarrhea and urine from individuals with urinary tract infections [6,7]. Additionally, *P. asiatica* has been recovered from a diverse range of sources, including environmental samples such as soil and water [13], human clinical specimens [14], and animal-related sources such as faeces [3]. Its presence across different ecological niches underscores its remarkable adaptability and potential for widespread dissemination. However, despite its emerging significance, comprehensive studies on the pathogenic mechanisms, host-pathogen interactions, and epidemiological patterns of *P. asiatica* remain limited. Given the well-established role of other *Pseudomonas* spp. in both human and animal infections, their documented pathogenicity, and their ability to acquire AMR traits, an in-depth investigation into *P. asiatica* is essential. Understanding its virulence factors, resistance mechanisms, and potential zoonotic transmission routes will be crucial for assessing its public health implications and developing effective control strategies. Although *P. asiatica* has been isolated from human patients, its role in bovine mastitis remains unexplored. We hypothesized that *P. asiatica* is an emerging etiological agent of clinical mastitis in dairy cows, possessing distinct virulence determinants and multidrug resistance profiles that may contribute to its pathogenicity, persistence in the farm environment, and therapeutic challenges. This study provides the first comprehensive analysis of *P. asiatica* strains isolated from milk, faeces, and farm soils of dairy cows diagnosed with CM, examining their prevalence, antibiogram profiles, resistance and virulence mechanisms. We performed WGS on four MDR *P. asiatica* strains [3], revealing their genomic relatedness to other *Pseudomonas* spp., along with the presence of ARGs and VGs. By elucidating the genetic determinants of resistance and virulence, this study lays a crucial foundation for the development of targeted therapeutic strategies to combat mastitis, promoting sustainable udder health management and ensuring the long-term viability of the dairy industry.

2. Materials and methods

2.1. Isolation, identification and antibiogram profiling of *P. asiatica* isolates

Milk, faeces, and soil samples were collected from cows diagnosed with CM using the California Mastitis Test [15] across 30 small-holding dairy farms (5–20 cows per farm) in Gazipur, Bangladesh (24.09° N, 90.41° E). The study strictly adhered to ethical guidelines for animal welfare during sample collection. The procedures were performed with minimal stress to the animals, following aseptic protocols to ensure hygiene and prevent discomfort. Prior to milk sampling, teat ends were disinfected to reduce

infection risks. All samples (milk, faeces, and soil) were collected non-invasively, with no harm or undue stress inflicted on the cows. While formal ethical approval was not mandated for this study (as per local regulations in Bangladesh for non-invasive sampling), we followed internationally recognized standards for animal welfare, including the OIE Terrestrial Animal Health Code and FAO guidelines for responsible antimicrobial use in livestock. Farmers provided verbal consent for sample collection, and no additional interventions beyond routine farm practices were employed. Finally, a total of 110 samples were collected, including 45 samples from CM milk, 35 from faeces, and 30 from farm soil. During the morning milking session (8:00–10:00 a.m.), approximately 10 mL of milk and 5 g each of faeces and soil were aseptically collected from each cow and farm into sterile Falcon tubes. Careful disinfection of the teat ends was performed prior to sampling, and strict adherence to hygiene protocols was maintained throughout the process to ensure sample integrity and minimize contamination. To isolate and phenotypically identify *P. asiatica*, samples were first enriched in nutrient broth (Oxoid, UK) incubated overnight at 37 °C. The bacteria-enriched broth was then serially diluted up to 10–3 fold and plated on nutrient agar (Oxoid, UK), followed by overnight incubation at 37 °C. Pure colonies were obtained by streaking on selective tryptic soy agar (TSA) (Oxoid, UK) plates by incubating at 37 °C for 24 h [16]. Phenotypic identification of the isolates was performed by examining colony morphology and Gram staining (Gram-negative, rod shape, pink colony) [14,17] while species-level identification was performed through a VITEK-2 system v9.01 [18] and ribosomal gene (16S rRNA) sequencing using UN18f 5'-AGAGTTTGATCCTGGCTCAG-3' and UN1492r 5'-GGTACCTGTACGACTT-3' primers [13]. Antimicrobial susceptibility patterns of the *P. asiatica* isolates ($n = 41$) were examined using the Kirby-Bauer disc diffusion method [19] following the guidelines of the Clinical Laboratory Standards Institute (CLSI) M100 33rd Edition [20] against 15 antibiotics (Table S1) that are commonly used to treat human and animal diseases including mastitis in Bangladesh. Antibiotic resistance breakpoints were defined according to CLSI guidelines [20]. The MDR *P. asiatica* isolates were defined as those exhibiting resistance to ≥ 3 antibiotics.

2.2. DNA extraction, whole genome sequencing, assembly and annotations of *P. asiatica* isolates

Genomic DNA from four selected MDR (resistant to > 8 antibiotics) isolates (e.g., 2M1, 2F1, 2F2 and 2S1; Table S2) were extracted using QIAamp DNA Mini Kit (QIAGEN, Hilden, Germany). Whole genome libraries were prepared using Nextera™ DNA Flex Kit (Illumina, USA) from 1 ng DNA, followed by WGS on an Illumina MiSeq with a 2×250 protocol [2,3]. Generated raw reads were trimmed using Trimmomatic v0.39 (with parameters leading: 20, slidingwindow:4:20:20, trailing:20, and minlen=36) [21], and quality checked using FastQC v0.11.7 [22]. The trimmed reads were assembled with SPAdes v3.15.5 [23]. The NCBI Prokaryotic Genomes Annotation Pipeline (PGAP) (https://www.ncbi.nlm.nih.gov/refseq/annotation_prok/) was used to annotate the assembled genomes. Furthermore, Rapid Annotation using Subsystem Technology (RAST, <https://rast.nmpdr.org/rast.cgi>) was used to identify number of subsystems present in each genome, while GenoVi v0.2.16 [24] was used for circular visualisation of the genomes.

2.3. Phylogenetic and average nucleotide identity (ANI) analysis of *P. asiatica* strains

The assembled *P. asiatica* genomes (2M1, 2F1, 2F2, and 2S1) were used to perform an evolutionary phylogenetic analysis alongside seventeen closely related *Pseudomonas* genomes (Table S3) available in the National Centre for Biotechnology Information

(NCBI) database (<https://www.ncbi.nlm.nih.gov/>). A maximum likelihood phylogenetic tree of these genomes was generated using the Type Strain Genome Server (TYGS; <https://tygs.dsmz.de/>) [25], and visualized in interactive tree of life (iTOL) v3.5.4 (<http://itol.embl.de/>). In addition, ANI was calculated to assess the overall similarity between the *Pseudomonas* genome sequences. ANI analysis was performed for all-against-all microbial genomes using ANIclustmap v1.3.0 [26].

2.4. Prediction of antimicrobial resistance and virulence genes in *P. asiatica* strains

The ARGs and VGs in the draft genomes were predicted using ABRicate v1.0.1 (<https://github.com/tseemann/abrigate>), which is integrated with multiple databases. ARGs were predicted in the assembled genome using NCBI AMR FinderPlus, CARD, and ResFinder. VGs across all study genomes were identified using the VFDB database, also incorporated within ABRicate. Additionally, genome annotation files were examined to identify genes associated with ARGs and VGs. Data were entered into Microsoft Excel 2019 (Microsoft Corp., Redmond, WA, USA) and analysed accordingly.

2.5. Statistical analysis

Data were recorded and analysed using Microsoft Excel 2020. The AMR patterns, including resistance, intermediate, and sensitivity, were calculated following the CLSI (2023) guidelines, using the cut-off values provided in the manufacturer's brochure (Oxoid, Hampshire, UK). AST data were analysed with one-way ANOVA, followed by Tukey's post-hoc test. ARGs and VGs data were normalized using total sum scaling. A p-value of ≤ 0.05 was considered statistically significant.

3. Results and discussion

3.1. Isolation, identification, and prevalence of *P. asiatica*

Forty-one ($n = 41$) isolates of *P. asiatica* were screened from 110 samples using conventional culture techniques and Gram stain-

ing, comprising 12 isolates from milk, 14 from faeces, and 14 from soil. On TSA agar, the isolates formed circular, smooth, moist, medium-sized, pale yellow to colorless, translucent, and slightly mucoid colonies. Microscopic analysis confirmed Gram-negative, pink-colored, rod-shaped bacteria. Species-level identification was confirmed through the VITEK-2 system and 16S rRNA-gene sequencing. The overall prevalence of *P. asiatica* was 37.27%, with the highest occurrence in soil (36.59%), followed by faeces (34.15%) and milk (29.27%). Previous studies from countries such as India, China, Brazil, and Egypt have reported the isolation of *Pseudomonas* species from bovine mastitis cases, often highlighting their multidrug-resistant (MDR) nature [27–29]. However, *P. asiatica* remains poorly characterized globally, with only a few reports of its presence in clinical settings, primarily in human infections. Its identification in bovine mastitis in Bangladesh marks a significant and novel finding, highlighting potential zoonotic risks and alarming resistance to frontline antibiotics. Previous studies have reported *Pseudomonas* species in dairy environments. Meng et al. [27] identified 143 *Pseudomonas* strains from bulk tank milk samples in China. Similarly, Aziz et al. [28] reported a 29.6% prevalence of *P. aeruginosa* in dairy farms in Egypt, with notable detection in milk (70.0%), milking machine swabs (32.0%), hand swabs (28.0%), and milk tanks (24.0%). Additionally, De Longhi et al. [29] documented the widespread presence of *P. putida* and *P. fluorescens* in milk-producing regions of Brazil, exhibiting 100% genetic similarity across locations. As a recently classified member of the *P. putida* group, *P. asiatica* is recognized as an emerging nosocomial pathogen. However, till now only 10 clinical isolates, including type strains, have been reported in Japan and Myanmar [6,7], emphasising the significance of our study in expanding the known ecological and clinical distribution of this species. Therefore, the emergence of *P. asiatica* in livestock introduces new challenges, given its phylogenetic ties to the *P. putida* group and its potential to disseminate MDR traits at the animal-human-environment interface.

3.2. Antibigram profiles of the *P. asiatica* isolates

In vitro antibiotic susceptibility testing of 41 *P. asiatica* isolates revealed that all were MDR, exhibiting resistance to at least

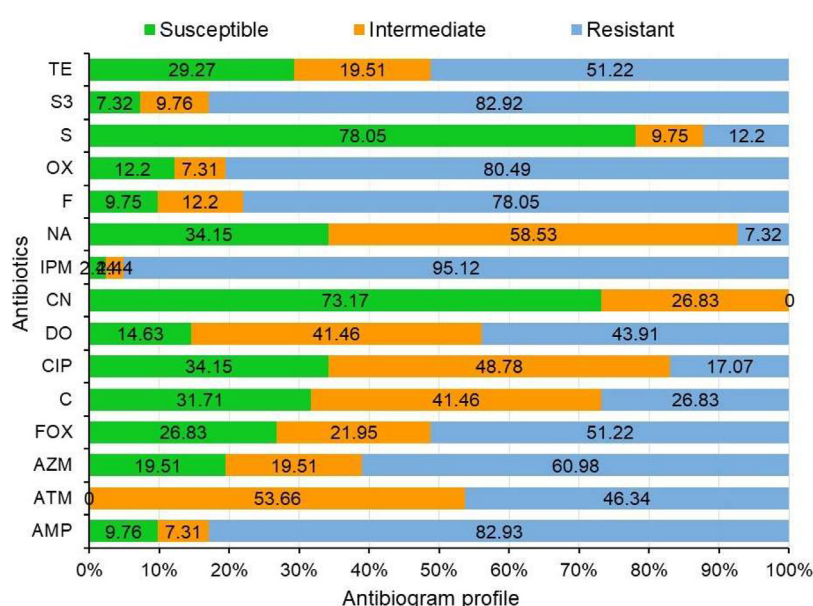


Fig. 1. Overall antibiotic resistance patterns of *P. asiatica* isolates ($n = 41$). Legends: AMP = ampicillin, ATM = aztreonam, AZM = azithromycin, FOX = cefoxitin, C = chloramphenicol, CIP = ciprofloxacin, DO = doxycycline, CN = gentamicin, IPM = imipenem, NA = nalidixic acid, F = nitrofurantoin, OX = oxacillin, S = streptomycin, S3 = sulfonamide, TE = tetracycline.

three antibiotics, with some resistant to up to 12. The highest resistance was observed against imipenem (95.12%), followed by ampicillin (82.93%), sulfonamides (82.92%), and oxacillin (80.49%) (Fig. 1). Additionally, 70–80% of isolates were resistant to nitrofurantoin, azithromycin, tetracycline, and cefoxitin, while moderate resistance was noted for doxycycline (43.9%) and aztreonam (46.3%). Lower resistance levels were recorded for chloramphenicol (26.83%), ciprofloxacin (17.07%), streptomycin (12.20%), and nalidixic acid (7.32%). However, *P. asiatica* remained notably sus-

ceptible (70–80%) to gentamicin and streptomycin, with moderate susceptibility (30–40%) to ciprofloxacin, nalidixic acid, and chloramphenicol (Fig. 1). Several previous studies reported that AMR emergence in mastitis-causing bacteria varies by bacterial type, origin, and host characteristics [5,8,16,30]. The antibiogram analysis of *P. asiatica* isolates screened from milk, faeces and farm soils underscored their clinical significance, as all isolates exhibited MDR to most tested antibiotics, suggesting a high potential for widespread dissemination. Such resistance poses a dual

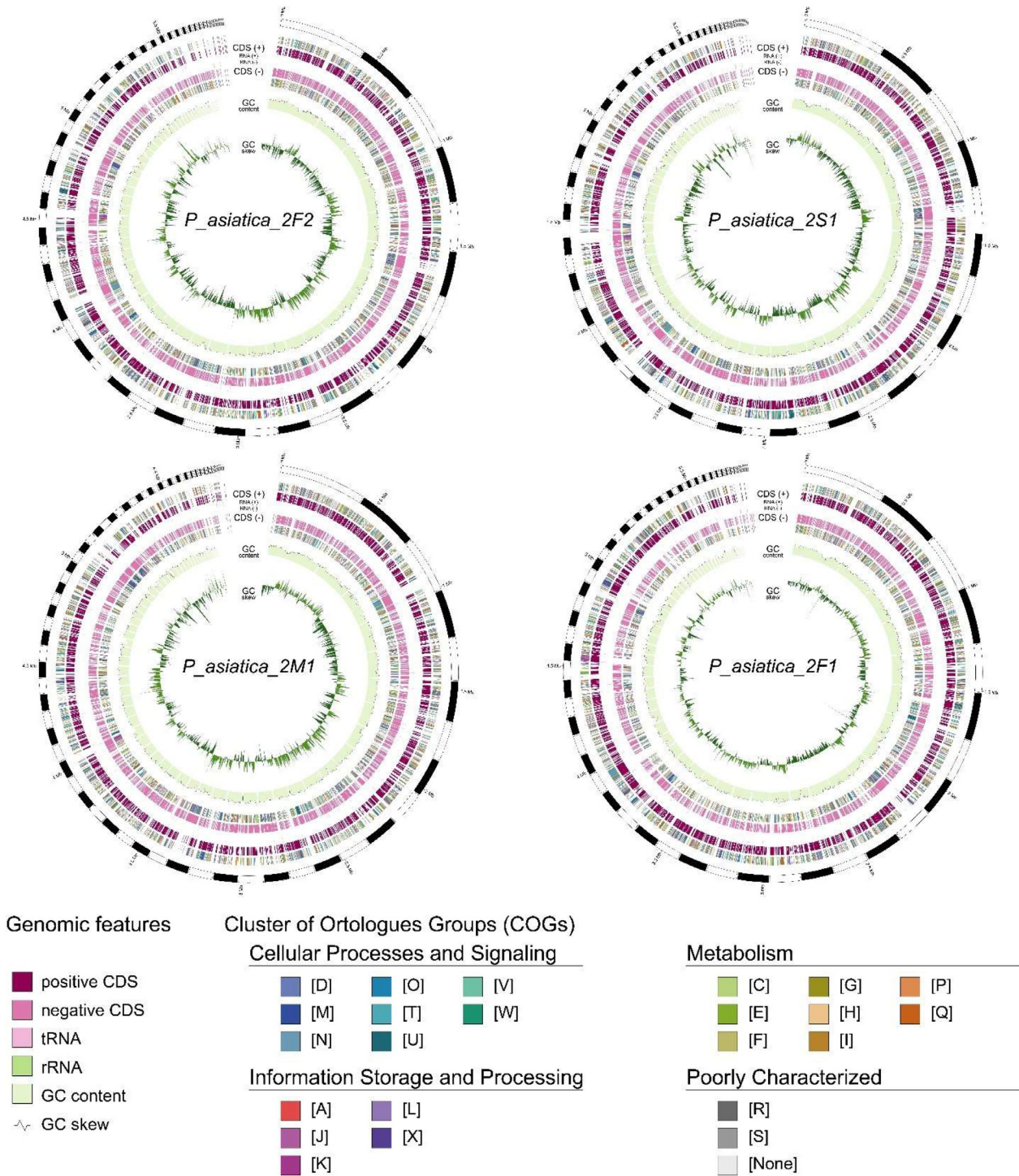


Fig. 2. Circular genome map with genomic features of the *P. asiatica* strains 2F2, 2S1, 2M1 and 2F1. Circular genome map of *P. asiatica* strains depicting from the outside to the center: genome contigs, coding sequences (CDS) on forward strand with COG category annotation, tRNA and rRNA, CDS on reverse strand with COG category annotation, GC content, and GC skew.

Table 1
General genomic features of the *P. asiatica* strains.

Features	<i>P. asiatica</i> strains			
	2M1	2F1	2F2	2S1
Genome size (bp)	5,672,760	5,663,297	5,660,490	5,665,922
Genome coverage (x)	50x	50x	50x	50x
GC content (%)	62.8	62.8	62.8	62.8
Total contigs	80	75	75	80
Largest contig (bp)	429,215	405,006	429,115	405,089
Shortest contig (bp)	1295	1268	1050	532
Contig N_{50} (bp)	122,302	150,834	150,834	150,767
L_{50}	14	13	13	13
Total genes	5264	5236	5239	5239
Coding sequences (CDSs)	5188	5170	5174	5173
Protein coding genes	5132	5116	5120	5119
rRNA genes	76	66	65	66
tRNA genes	66	58	57	58
rRNAs	4, 1, 1 (5S, 16S, 23S)	2, 1, 1 (5S, 16S, 23S)	2, 1, 1 (5S, 16S, 23S)	2, 1, 1 (5S, 16S, 23S)
ncRNAs	4	4	4	4
Pseudo genes	56	54	54	54
Number of subsystems	371	371	371	369
Number of prophages	2	2	2	2
No. of ARGs	11	11	11	11
No. of VGs	30	30	30	30
16S rRNA gene accessions	PV123076	PV123077	PV123078	PV123079
Genome accessions	JAXUBP000000000	JAXUBP000000000	JAXUBP000000000	JAXUBP000000000
SRA accessions	SRR26993013	SRR26993013	SRR26993013	SRR26993013

threat by limiting treatment options for animal infections and facilitating the transfer of resistance genes to humans, increasing the global AMR crisis [31]. While gentamicin and streptomycin remain effective against *P. asiatica*, cautious use is vital to mitigate AMR and protect animal health, human health, and food security. In dairy farming, the presence of AMR in pathogens like *P. asiatica* can cause significant economic losses, while transferring AMR to human cause serious health problems. If this trend continues, it is projected that AMR could surpass all other causes of mortality worldwide, presenting a critical global health challenge [12].

3.3. Genomic features and phylogenetic relationship of the *P. asiatica* strains

In this study, four MDR *P. asiatica* isolates including one from milk (2M1), two from faeces (2F1, 2F2), and one from soil (2S1) were selected for WGS. The assembled genomes averaged 5.7 Mbp in size, with a GC content of 62.8% and a sequencing coverage of 50x (Fig. 2). Detailed information about these isolates is provided in Table 1. Notably, subsystem annotation using the RAST server revealed that the strains isolated from milk and faeces (2M1, 2F1, and 2F2) possessed 371 annotated subsystems, whereas the

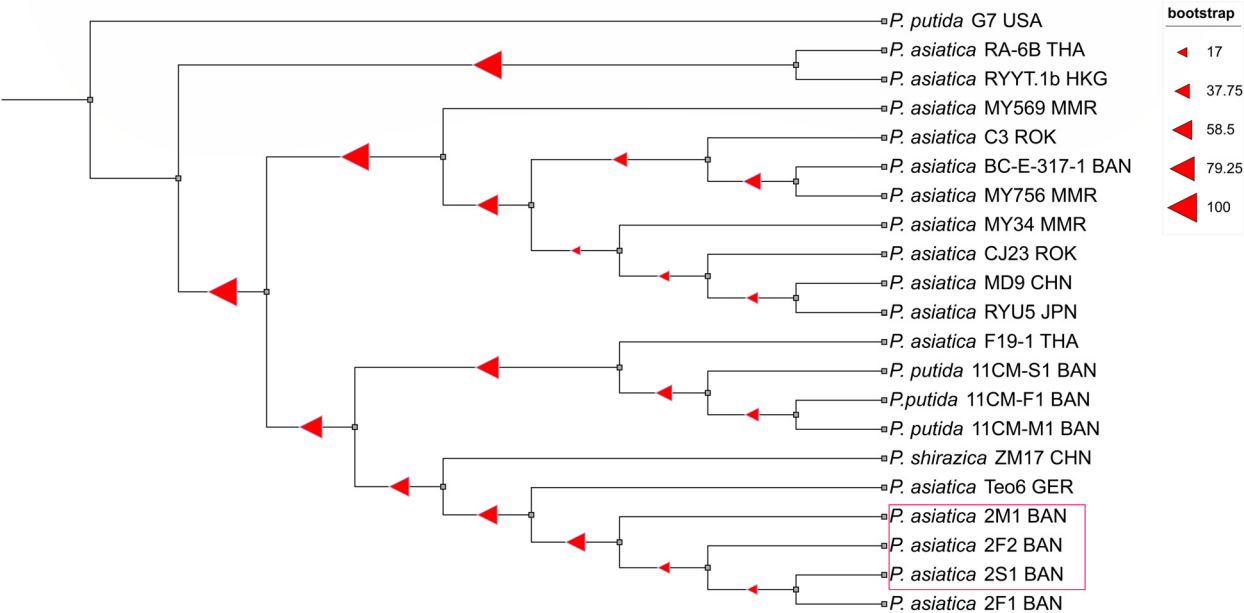


Fig. 3. Phylogenomic tree built using the TYGS (<https://tygs.dsmz.de>), for taxonomic analysis based on the genome of closely related strains [25]. Tree inferred with FastME 2.1.6.1 (<http://gensoft.pasteur.fr/docs/fastme/2.1.6.1>) from Genome Blast Distance Phylogeny (GBDP) distances calculated from genome sequences. The branch lengths are scaled in terms of GBDP distance formula d5. The numbers above branches are GBDP pseudo-bootstrap support values > 60% from 100 replications.

soil isolate (2S1) exhibited 269 subsystems. Phylogenetic analysis revealed that the studied genomes clustered closely with *P. shirazica* ZM17 (soil, CP075569.1), *P. asiatica* Teo6 (activated sludge, NZ_CP139778.1), *P. putida* strains 11CM-M1 (mastitis milk), 11CM-F1 (faeces of a mastitic cow), and 11CM-S1 (soil from a mastitis-affected farm) [32], as well as *P. asiatica* F-19 (environmental sample from pig farms, NZ_CP128508.1) (Fig. 3). These six closely re-

lated *Pseudomonas* genomes were isolated from Bangladesh, China, Germany, and Thailand. This close relationship is particularly significant, as it indicates that MDR *P. asiatica* may share common reservoirs or evolutionary origins with other pathogenic or environmentally adaptable *Pseudomonas* strains [33]. The geographic distribution of these related genomes across diverse regions also supports the notion of a widespread and potentially mobile re-

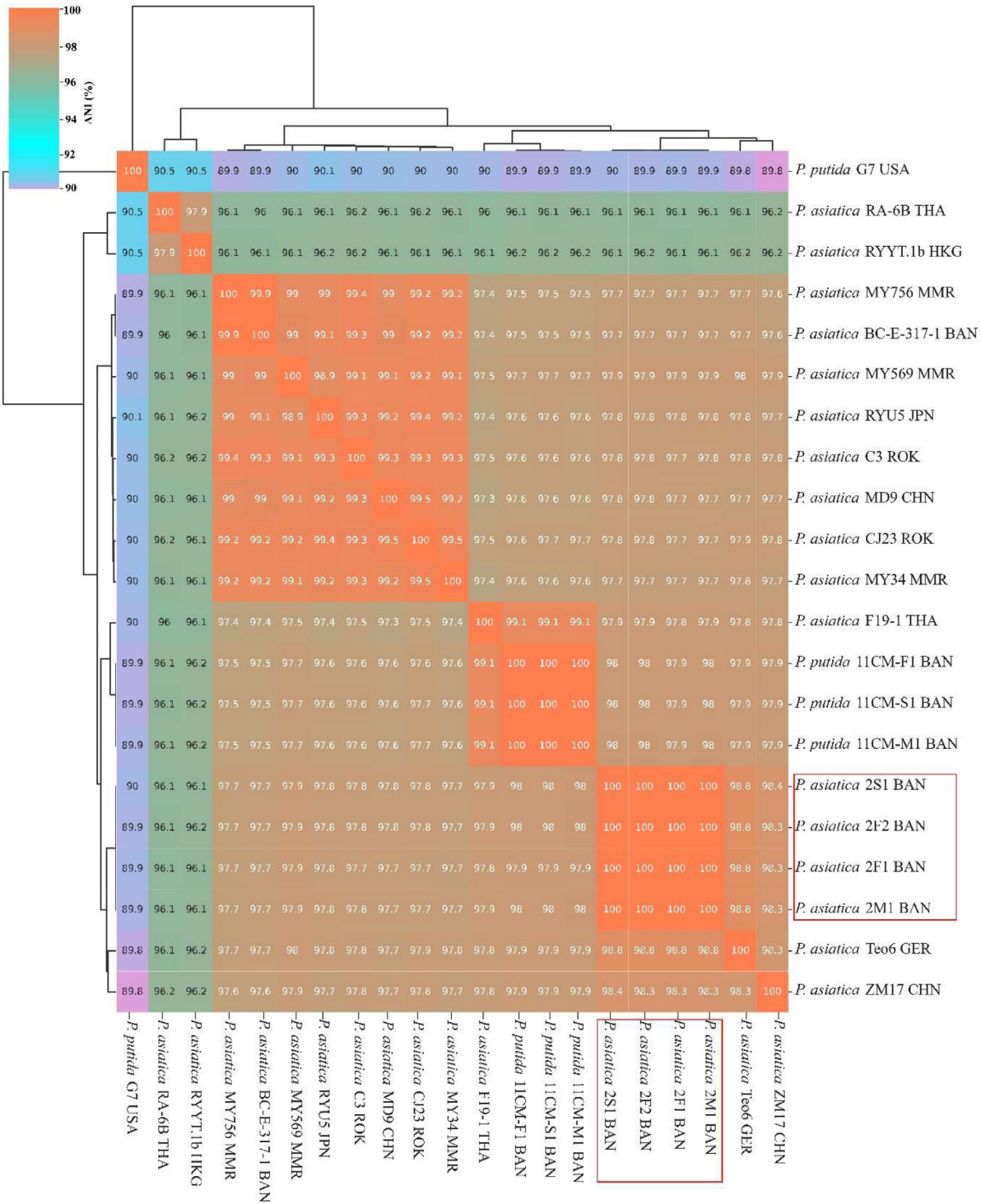


Fig. 4. Clustered heatmap of average nucleotide identity (ANI) values among study genomes (*P. asiatica* 2M1, 2F1, 2F2 and 2S1) and 17 reference genomes of different strains of *Pseudomonas* spp. (previously used in phylogenetic analysis). The numbers represent the ANI values (%) between two genome sequences. An ANI above 95% between two genomes is an indication that they belong to the same species. The color codes indicate ANI values, yellow for 100% identity and whitish purple for 80.0% identity. Study genomes are highlighted in black box (in the Y-axis).

sistome within the *Pseudomonas* genus. The findings of the phylogenetic analysis were further supported by ANI analysis, revealing ANI scores of >97% between the closely related genomes and 100% within our sequenced genomes (Fig. 4). These phylogenetic and ANI results indicated high genomic conservation within the studied strains and significant species-level similarity with other isolates from diverse sources and geographical locations [34]. This suggests extensive genome adaptation and high genetic diversity among *P. asiatica* strains.

3.4. Resistome and virulome repertoire of the *P. asiatica* strains

One of the key findings of this study was the identification of 11 ARGs in the genomes of *P. asiatica* strains 2M1, 2F1, 2F2, and 2S1, including *mexB*, *mexE*, *mexF*, *mexK*, *mexI*, *mexW*, *ttgA*, *ttgB*, *ttgC*, *ttgR*, and *CpxR*. By comparing the mechanism of resistance of the predicted ARGs, we found that > 80% genes were encoding for antibiotic efflux pump, whereas, rest of the genes encode for drug and biocide resistance regulator and multi-compound resistance though efflux pump. The identified genes *mexB*, *mexE*, *mexF*, *mexK*, *mexI*, and *mexW* are all integral components of the Mex efflux pump system, which plays a crucial role in antibiotic resistance by facilitating the expulsion of antimicrobial agents from bacterial cells. Mex efflux pumps in *P. aeruginosa* contribute to MDR by pumping out a wide range of antibiotics (e.g., fluoroquinolones, tetracyclines, macrolides, aminoglycosides, carbapenems, cephalosporins, and sulfonamides), disinfectants, and tox-

ins [35]. The *mexB* gene eject active molecules of antibiotic toward the periplasmic tunnel and outer membrane in coordination with other pump-associated proteins making bacteria resistance to antibiotic [35]. We also identified *mexE* and *mexF* as crucial components of the MexEF-OprN efflux pump, which has been shown to confer resistance to antimicrobial agents, including fluoroquinolones, trimethoprim, and chloramphenicol. [36]. Notably, we predicted the *oprM* gene in all of the analysed genomes, with locus tags SOW74_RS11395 (2F1), SOW75_RS10410 (2F2), SOW72_RS19210 (2M1), and SOW73_RS11765 (2S1). MexJK, a variant of the Mex efflux pump, plays a crucial role in bacterial resistance by facilitating the extrusion of antimicrobial agents from the cell [37]. The widespread presence of the *mexK* and *oprM* genes in all *P. asiatica* strains supports the development of multidrug resistance in our isolated species. Furthermore, the *mexI* and *OprD* genes (locus tags: SOW74_RS02940 [2F1], SOW75_RS00600 [2F2], SOW72_RS01420 [2M1], and SOW73_RS01310 [2S1]) have been identified in all sequenced *P. asiatica* genomes.

Each of the *P. asiatica* strains (2M1, 2F2, 2F2, and 2S1) was found to possess a set of 30 VGs (Fig. 5A). The most important VGs predicted in *P. asiatica* strains include genes associated with flagellar/twitching motility activities (e.g., *flhA*, *fliG/I/M/N/P/Q*, *flgC/H*, *fleN*, *pilH*, *motC*; ~ 40.0%), alginate biosynthesis and regulation (e.g., *algG*, *algC/U/W*; ~ 21.0%), two-component regulatory system (e.g., *algB*; ~ 9.0%), transcriptional regulator (e.g., *fleQ*; ~ 7.0%), heptosyltransferase (e.g., *waaF*; ~ 6.0%), phosphomannose isomerase (e.g., *algA*; ~ 5.0%), extracytoplasmic-function sigma-70 factor (e.g.,

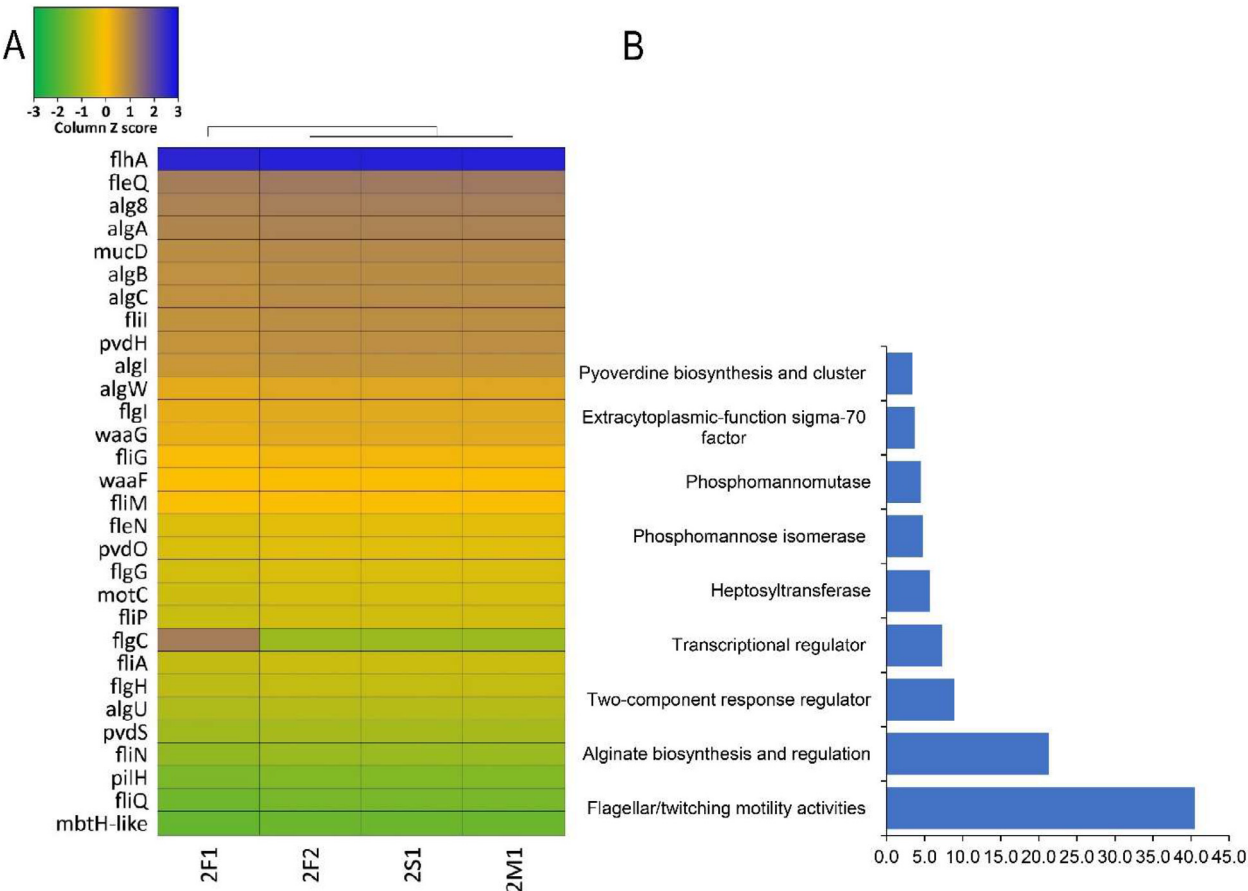


Fig. 5. Virulome of *P. asiatica* strains. (A) Clustered heatmap of 30 VGs based on coverage, number of hits and identity. The color bar (row Z score) at the top represents the relative abundances of the respective VGs *P. asiatica* genomes. The color codes indicated the presence and completeness of each VG, expressed as a value between -3 (lowest abundance) and 3 (highest abundance). The blue color indicates the highest abundant patterns, while green cells account for the least abundant VGs. The heatmap is generated through FunRich (<http://www.funrich.org/>). (B) Bar plot representing the mechanisms of virulence in *P. asiatica* strains by the detected VGs. The Y-axis represents the mechanisms of virulence while their relative abundances (%) are shown in X-axis.

pvdS; 4.0%) and pyoverdine biosynthesis and cluster (e.g., *pvdO*, *mbtH*; ~ 3.5%) (Fig. 5B). The flagellar/twitching activities are crucial for bacterial motility, chemotaxis, and pathogenicity, aiding in environmental adaptation, biofilm formation, immune modulation, and tissue adhesion/invasion [38]. Genes encoding flagellar biosynthesis proteins (e.g., *flhA*, *fliA*, *fliQ*, *fliP*) are crucial for flagella synthesis and have also been implicated in bacterial pathogenesis [39]. One of recent studies reported that deletion of the *fliP* gene markedly diminished the virulence of *Pseudomonas plecoglossicida*. [40]. Additionally, *fleQ* and *fleN* reported as master regulators of both flagellar motility and biofilm formation in *P. aeruginosa*, that significantly regulate gene expression and pathogenesis [41]. The *algA*, *algB*, *algC*, *algI*, *algU*, *alg8*, and *mucD* genes were identified in the studied bacteria are primarily involved in the precursor formation and/or synthesis of alginate, and play vital as main virulence factor in *P. aeruginosa*. The *algB* gene regulates alginate production and two-component signal transduction systems (TCSTS) in *P. aeruginosa* [42]. TCSTS are crucial for *Pseudomonas* survival and growth, facilitating virulence factor expression and antibiotic resistance development [43]. Moreover, *algC* is a crucial gene required for the biosynthesis of three major virulence factors alginate, rhamnolipids, and lipopolysaccharides in *P. aeruginosa* [44]. The *algI* gene enhances biofilm formation and helps *P. aeruginosa* evade phagocytosis through the development of a mucoid phenotype [45]. In addition, pyoverdine-related genes (*pvdH*, *pvdS*, *pvdO*, and *mbtH*-like) were found in all *P. asiatica* strains. Pyoverdine aids iron acquisition, biofilm formation, and immune evasion [46]. The *wwF* and *wwG* genes contribute to lipopolysaccharide biosynthesis, vital for immune evasion and pathogenicity [47]. The *pilH* gene supports type IV pilus biogenesis, motility, and biofilm formation [48]. Furthermore, the genomes of *P. asiatica* harbor genes encoding the Type VI Secretion System (T6SS), including *tssABCEFGHIJKM* and *vgrG*. This sophisticated system plays a crucial role in bacterial pathogenicity by promoting interbacterial competition, biofilm development, and adaptation to environmental challenges, ultimately contributing to pathogenesis [49]. Notably, quorum-sensing-regulated virulence factor family proteins were also identified in the study genomes which may allow the bacteria to evade host immune defenses and establish infection [50]. Although the pathogenicity of *P. asiatica* in mastitis remains undocumented, *Pseudomonas* species generally possess pathogenic traits that can cause infections in humans, animals, and plants, with pathogenicity varying among strains. The challenges in managing *Pseudomonas*-induced mastitis include limited awareness, diagnostic difficulties, and high resistance to multiple antimicrobials, often rendering conventional treatments ineffective. Additionally, *Pseudomonas* can persist in the farm environment, increasing the risk of reinfection and complicating control measures.

In summary, this is the first WGS-based study on MDR *P. asiatica* strains 2M1, 2F1, 2F2, and 2S1 isolated from dairy environments in Bangladesh. This study provides insights into the AMR profile, phylogenetic relationships, ARGs, and VGs in *P. asiatica* strains isolated from mastitic dairy environments (including milk, faeces and farm soils), offering critical insights into the dissemination mechanisms of resistance and virulence among bacteria, animals, and humans. The *P. asiatica* strains isolated from dairy environments are valuable for global databases, particularly for their clinical relevance to mastitis and human infections. The emergence of AMR in *P. asiatica* isolates poses a public health risk, potentially transferring AMR traits to humans. This study identified numerous VGs in *P. asiatica* strains from mastitis-associated dairy farms, which contribute to pathogenicity, biofilm formation, immune evasion, host interaction, and MDR, underscoring the importance of *P. asiatica* in mastitis and its impact on the dairy industry of Bangladesh. These findings highlight the need for further investigation of mastitis-associated *P. asiatica* strains to understand their pathogenic mech-

anisms and zoonotic potential. While this is the first study linking *P. asiatica* to mastitis, its known clinical significance in humans stresses the urgency of targeted interventions for animal health, public health, and the dairy industry.

Data availability

The whole-genome shotgun project of the *P. asiatica* strains has been deposited in GenBank and the NCBI Sequence Read Archive (SRA) under BioProject accession number [PRJNA1046646](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1046646). The SRA accessions are listed in Table 1. The version described in this paper are versions [JAXUBP000000000.1](https://www.ncbi.nlm.nih.gov/assembly/JAXUBP000000000.1), [JAXUBN000000000.1](https://www.ncbi.nlm.nih.gov/assembly/JAXUBN000000000.1), [JAXUBM000000000.1](https://www.ncbi.nlm.nih.gov/assembly/JAXUBM000000000.1), and [JAXUBO000000000.1](https://www.ncbi.nlm.nih.gov/assembly/JAXUBO000000000.1). The 16S rRNA data of the four multidrug-resistant *P. asiatica* strains 2M1, 2F1, 2F2 and 2S1 are available at NCBI GenBank under accessions PV123076, PV123077, PV123078 and PV123079, respectively.

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Declaration of competing interests

None declared

Ethical approval

This study was approved by the Animal Research Ethics Committee (AREC) of Gazipur Agricultural University, Bangladesh, with reference number FVMAS/AREC/2024/6755, granted on January 20, 2024. Additionally, none of the authors conducted studies involving human participants.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jgar.2025.05.022](https://doi.org/10.1016/j.jgar.2025.05.022).

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